

CS5326 – Advanced GenAI and Agents
Fall 2026

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Long-Context Extension



The Central Question

How do we fit a **longer sequence** into a model's positional experience while **preserving useful distinctions between nearby tokens**?

A formula can outgrow its training

Defined at a new position does not mean learned behavior generalizes there.



Suppose we have already trained our model using RoPE with a context length of 4,000, but we want to support a context length of 12,000 for inference.

Let's start with four rotating hands of RoPE

- Assume

Symbol	Meaning	Our example
L	Original training context length	4,000 tokens
L'	Desired context length	16,000 tokens
$s = L' / L$	Context extension factor	4
d	Query/key dimension in one head, all rotated here	8
b	RoPE's frequency base	10,000
i	Coordinate-pair index	0,1,2,3

- An eight-dimensional query has four pairs:

$$q = \left(\underbrace{q_0, q_1}_{i=0}, \underbrace{q_2, q_3}_{i=1}, \underbrace{q_4, q_5}_{i=2}, \underbrace{q_6, q_7}_{i=3} \right)$$

- Each pair gets its own angular speed:

$$\omega_i = b^{-2i/d}$$

- That is** “Start with speed 1, then assign progressively smaller speeds to later pairs.”

Setting up Conventions

An eight-dimensional query has four pairs:

$$q = \left(\underbrace{q_0, q_1}_{i=0}, \underbrace{q_2, q_3}_{i=1}, \underbrace{q_4, q_5}_{i=2}, \underbrace{q_6, q_7}_{i=3} \right)$$

- Each pair gets its own angular speed:

$$\omega_i = b^{-2i/d}.$$

For $d = 8$:

$$\omega_i = 10000^{-i/4}$$

Because $10000 = 10^4$:

$$10000^{-i/4} = (10^4)^{-i/4} = 10^{-i}$$

So our four speeds are particularly simple:

Pair i	Angular speed ω_i , radians/token	Angle at position p = 10: $p\omega_i$
0	1	10 radians
1	0.1	1 radian
2	0.01	0.1 radians
3	0.001	0.01 radians

The angle formula is simply:

$$\underbrace{\phi_i(p)}_{\text{angle}} = \underbrace{p}_{\text{token position}} \underbrace{\omega_i}_{\text{radians per token}}$$

It is the rotational equivalent of **distance = time (s) × speed (m/s)**.

- For example, at position 10, a hand moving 0.1 radians/token has turned $10(0.1) = 1$ radian.

Why does relative position appear in RoPE?

For **one coordinate pair**, suppose the query and key initially point at angles a and b . RoPE adds their positional rotations:

$$\text{Query angle} = a + m\theta, \text{Key angle} = b + n\theta.$$

Their angular difference is therefore:

$$(a + m\theta) - (b + n\theta) = \underbrace{(m - n)\theta}_{\text{positional difference}} + \underbrace{(a - b)}_{\text{original content-angle difference}}.$$

Using **dot product = product of lengths $\times$ cosine of the angle between them**:

$$\boxed{\tilde{q}^\top \tilde{k} = \|q\| \|k\| \cos((m - n)\theta + (a - b))}$$

Using our example for RoPE

Query position: $m = 10$; **key position:** $n = 0$.

Original frequency: $\theta = 0.1$ radians/token.

Extension factor: $s = 4$, extending from 4,000 to 16,000 tokens.

Unrotated pairs: $q = k = (1, 0)$, so their lengths are 1 and initial angles are $a = b = 0$.

Original RoPE

The query and key rotate to:

$$\text{Query angle} = a + m\theta = 0 + 10(0.1) = 1,$$

$$\text{Key angle} = b + n\theta = 0 + 0(0.1) = 0.$$

Their dot product is:

$$\begin{aligned}\tilde{q}^\top \tilde{k} &= \|q\| \|k\| \cos((m - n)\theta + (a - b)) \\ &= 1(1) \cos((10 - 0)(0.1) + (0 - 0)) \\ &= \cos(1) \approx \boxed{0.5403}\end{aligned}$$

- This is the cosine of 1 radian.

Remember RoPE

Every token uses all four hands. These are four pairs within each Q/K vector. We'll just simplify below:

What does “rotate the pair” actually do?

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi \\ \sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Multiplying out:

$$x' = x \cos \phi - y \sin \phi, y' = x \sin \phi + y \cos \phi.$$

For the simple vector $(1, 0)$, this becomes:

$$(1, 0) \mapsto (\cos \phi, \sin \phi).$$

At $\phi = 1$:

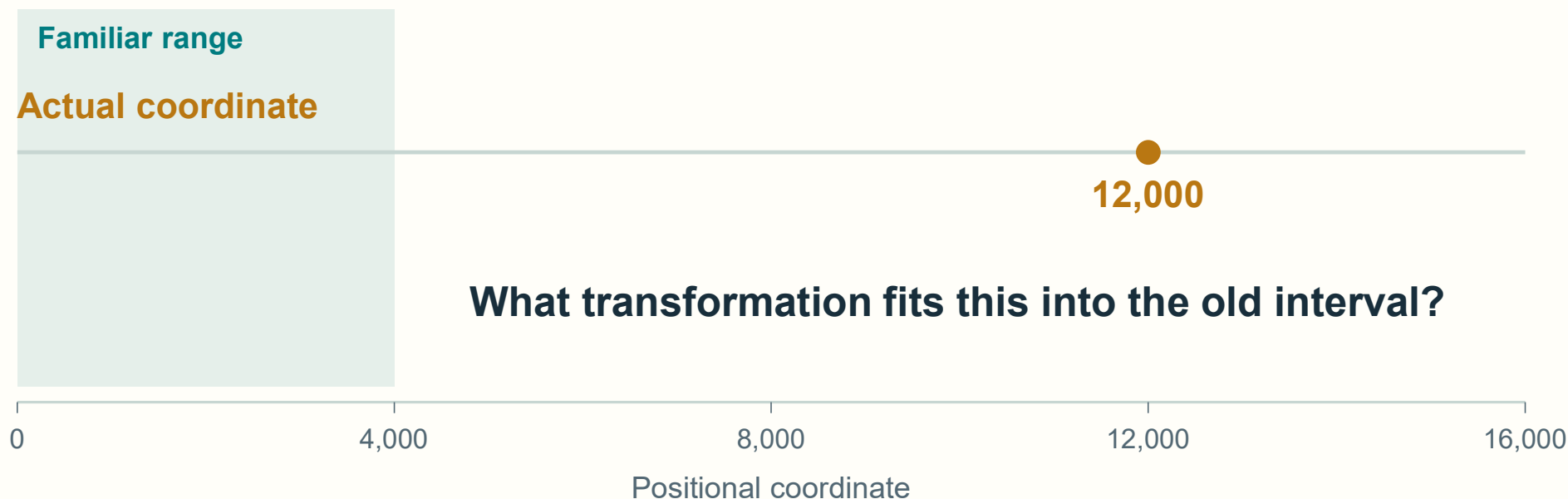
$$(1, 0) \mapsto (0.5403, 0.8415).$$

The arrow turns; its length remains 1.

POSITION INTERPOLATION

Fit the longer sequence onto the old scale

The target is 16,000 tokens. The familiar coordinate interval is 0–4,000.



Keep the tokens. Change the ruler used to assign their rotations.

Position Interpolation

The simplest fix: use $p/4$ in place of p .

$$\text{new angle} = \frac{p}{4} \omega_i$$

Thus, actual position 12,000 receives the rotation previously associated with position 3,000.

We still process all 16,000 tokens. We only compress their positional coordinates.

Equivalently, **slow every clock hand to one-quarter speed.**